AN UNIFIED AI-BASED NETWORK INTRUSION DETECTION FRAMEWORK

Functional Document

1. Introduction:

The AI-IDS-SOC (Artificial Intelligence – Intrusion Detection System – Security Operations Center) is a unified, end-to-end cybersecurity framework designed to detect and respond to network and host-level threats in real time.

The system addresses the following shortcomings of traditional IDS solutions:

- Signature-based systems are ineffective against zero-day exploits
- Basic ML models cannot handle mixed data streams from network and host layers
- Security staff are overwhelmed with excessive false alerts

In this framework, three orthogonal AI models are applied in cascade:

- Isolation Forest – Unsupervised anomaly detection
- Random Forest – Supervised flow classification
- Graph Neural Network (GNN) – Relational and multi-stage attack detection

A hybrid command-level detector further monitors CLI activity for fileless malware and Living-Off-The-Land (LOTL) attacks, all visualized through a real-time Flask-SocketIO dashboard.

2. Product Goal:

The main goal of this project is to:

- Detect both known and zero-day network attacks in real time
- Monitor command-line activity to identify fileless malware and LOTL attacks
- Achieve high True Positive Rate (TPR) with minimal False Positive Rate (FPR)
- Provide SOC analysts with a real-time, actionable dashboard
- Enable scalable deployment for small-to-medium SOC environments

Final aim: Build a production-ready, open-source AI-powered IDS/SOC system that reduces analyst alert fatigue and improves multi-layered threat detection coverage.

3. Demography (Users, Location):

Users:

- SOC Analysts (primary users monitoring real-time alerts)

Network Security Engineers (configure detection models and flow rules)

Cybersecurity Researchers (evaluate and extend the detection framework)

IT Administrators (manage system integration and infrastructure)

Students and Academic Researchers (study AI-based intrusion detection)

Location:

Can be deployed globally in enterprise and cloud environments

Useful in IT companies, academic research labs, and cybersecurity training centers

Particularly relevant for organizations managing high-volume API, web, and backend traffic

4. Business Processes:

Packet Capture Process

Scapy monitors network interfaces and captures raw packets in real time

■ Feature Engineering Process

Captured packets are converted into flow feature vectors:

Packet counts, byte counts, and directional features

Flow duration, protocol type, and statistical aggregates

■ Model Inference Process

Network flows are processed through three AI models in cascade:

Isolation Forest (anomaly scoring)

Random Forest (flow classification)

Graph Neural Network (relational pattern analysis)

■ Command Monitoring Process

CLI commands are monitored and passed through:

Regex rule engine (47 predefined patterns)

AI anomaly detector (Isolation Forest on command features)

■ Result Generation

System classifies each flow or command as:

Safe ■

Suspicious ■■

Malicious ■

5. Features:

■ Feature #1: Anomaly-Based Detection (Isolation Forest)

■ **Description:**

Detects zero-day and unseen attack patterns through unsupervised anomaly scoring on network flows, without requiring labeled training data.

■ **User Story:**

"As a SOC analyst, I want the system to flag unusual traffic patterns automatically so that zero-day threats are detected even without known signatures."

■ **Feature #2: Supervised Flow Classification (Random Forest)**

■ **Description:**

Classifies network flows against labeled historical data. Achieves 95.6% TPR and 2.6% FPR for known attack types including port scans, C2 beacons, and data exfiltration.

■ **User Story:**

"As a security engineer, I want the system to accurately classify known attack types so that analysts can quickly prioritize and respond to confirmed threats."

■ **Feature #3: Relational Attack Detection (Graph Neural Network)**

■ **Description:**

Builds a dynamic k-NN graph of network flows to detect multi-stage and lateral movement attacks through structural pattern analysis — threats that single-flow models typically miss.

■ **User Story:**

"As a SOC analyst, I want the system to detect coordinated and distributed attacks so that complex multi-stage intrusions are not missed."

■ **Feature #4: Hybrid Command-Level Detection**

■ **Description:**

Monitors CLI activity using 47 Regex rules combined with an AI-based Isolation Forest model to detect reverse shells, Base64-encoded payloads, and fileless attack techniques.

■ **User Story:**

"As a security analyst, I want the system to monitor shell commands in real time so that Living-Off-The-Land and fileless attacks are detected at the host level."

■ **Feature #5: Real-Time SOC Dashboard**

■ **Description:**

A Flask-SocketIO web dashboard displays live alerts, network flow graphs, model confidence scores, and historical event data for SOC analysts.

■ **User Story:**

"As a SOC analyst, I want a unified dashboard showing network and command alerts so that I can monitor and respond to threats efficiently from a single interface."

6. Authorization Matrix:

Define the roles and their corresponding access levels:

Role	Access Level
Administrator	Full access (manage system, models, rules, and SQLite database)
SOC Analyst	Access to live dashboard, alert acknowledgement, and historical data
Security Engineer	Access to update detection rules and retrain models
Student / Researcher	Can input test traffic and view model results
Guest User	Limited access (basic dashboard view only)

7. Assumptions:

- Network traffic is captured in real time using Scapy on monitored interfaces
- The labeled dataset of 200 flows (113 attacks, 87 benign) is representative of production traffic
- The system is deployed in a Linux environment with Python 3.x and required ML libraries installed
- SQLite is used for alert storage in the current version; PostgreSQL migration planned for enterprise scale
- Team members have the required knowledge of Python, machine learning, and basic networking concepts
- The Flask dashboard is accessible over the local network or VPN for authorized SOC personnel